

Alex Melendez

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EDUCATION

University of Houston | Bachelor of Science in Computer Science

Expected, December 2026

- GPA: 3.3/4.0 Major GPA: 3.6/4.0
- Relevant courses: Programming and Data Structures, Operating Systems, Database Systems, Algorithms and Data Structures, Data Science I, Software Design
- Organizations: Code Coogs, CS Girls, Latinas in Tech, Computing Alliance of Hispanic-Serving Institutions

SKILLS

Languages: C++, HTML, Java Script, CSS, MySQL, postgresql

Tools: Figma, Git/Github, Excel, Office, R, Adobe Suite, Postman

Soft Skills: Creativity, Problem Solving, Leadership, Verbal and Written Communication

Frameworks/Libraries: Express.js, React.js, Node.js

EXPERIENCE

Lead Sensei

Riverstone, TX

Code Ninjas

Feb. 2026 - Present

- Led and mentored a team of Senseis to deliver engaging coding and STEM instruction to students ages 5–14.
- Taught programming concepts including game development, robotics, and computer science fundamentals using age-appropriate instructional strategies.
- Developed engaging activities that encouraged computational thinking and collaborative problem-solving.

STEM CREW

Houston, TX

Girlstart

Aug. 2025 - June 2026

- Lead hands-on STEM-focused after school programs in Houston, specifically in Alief ISD for groups of 20-25 girls
- Organized and prepared class materials and registration forms for many groups of students

CAHSI LREU Participant

Houston, TX

University of Houston, under Professor Guoning Chen

Feb. 2025 - May 2025

- Developed an intuitive interface for the user to specify and modify different geometric and physical constraints for vector field synthesis
- Conducted mentored research as part of CAHSI local REU program
- Created a research plan, maintained a journal to report on research progress, and created a research poster to disseminate my research results
- Received CAHSI-Computing Research Association (CRA) Scholar Award

PROJECTS

Library Database | library-project-iota.vercel.app

- Developed a full-stack library web application that allows users to browse/checkout items working in a team of 4 as a class project.
- Built with **React** and a custom backend using **Native Node.js**, and **MySQL**
- Incorporated JWT-based Staff/User authentication layer to implement role-based access control
- Hosted the website using AWS

Volunteer Website | github.com/alexamelen/Volunteering_WebApp

- Developed a full-stack volunteer web application that allows users to browse/signup for events, working in a team of 4 as a class project.
- Built with **React** and used **Node.js (Express)** and **postgresql** to implement the back end
- Implemented the volunteer history mechanism and volunteer dashboard backend routing
- Hosted the website using AWS